

$0^0 = 1$ AND $0^x = 0$

The Thaumaturgical Computation

Michael Grafiel S Puno

$0^0 = 1$: The Creation Singularity

Standard mathematics leaves 0^0 undefined. But in four contexts, setting $0^0 = 1$ is not just convenient -- it is necessary:

COMBINATORICS: The number of functions from the empty set to the empty set is exactly 1 (the empty function). The void has ONE structure. $0^0 = 1$ means: the void is not nothing.

SET THEORY: $|\{\}|^{|\{\}|} = 1$. The empty set has cardinality 0, and the empty function is the unique function from $\{\}$ to $\{\}$. $0^0 = 1$ means: the empty set has STRUCTURE.

POWER SERIES: $\sum a_n x^n$ evaluated at $x=0$ gives $a_0 * 0^0$. If $0^0 = 1$, the constant term survives. The power series has a well-defined value at the origin.

EULER PRODUCT: At $s = 0$, $\zeta(0) = \prod_p 1/(1 - p^0)$. With $0^0 = 1$: $p^0 = 1$, so each factor is $1/(1-1) = 1/0 = \text{inf}$. The product DIVERGES. But $\zeta(0) = -1/2$ (finite!). The divergent product is regularized to $-1/2$.

$0^x = 0$ ($x > 0$): The Annihilation Singularity

ALGEBRA: $0^x = 0 * 0 * \dots * 0 = 0$. Zero multiplied by itself any number of times is zero. The origin ANNIHILATES everything.

CALCULUS: $\lim_{a \rightarrow 0} a^x = 0$ for $x > 0$. The function approaches zero. $0^x = 0$ means: the limit is ZERO, not undefined.

FRIEDMANN EQUATION: $a(t) = (3t/2)^{2/3}$. At $t = 0$: $a = 0^{2/3} = 0$. The scale factor VANISHES. The Big Bang singularity exists.

EULER PRODUCT ($\text{Re}(s) \gg 1$): For large $\text{Re}(s)$, $p^{-s} \rightarrow 0$. Each factor $1/(1 - p^{-s}) \rightarrow 1$. The product is trivial. Far from the critical line, zeta has no interesting structure.

The Transition: $0^0 = 1 \rightarrow 0^x = 0$

The transition from $0^0 = 1$ to $0^x = 0$ is the END OF INFLATION.

During inflation: $a(t) = \exp(Ht)$, $a(0) = 1 = 0^0$. The universe has FINITE size at $t = 0$. No singularity.

After inflation: $a(t) = (3t/2)^{2/3}$, $a(0) = 0 = 0^{2/3}$. The universe has ZERO size at $t = 0$. Big Bang exists.

At the transition t_{reheat} , both expressions give the SAME value:
 $\exp(H * t_{\text{reheat}}) = (3 * t_{\text{reheat}} / 2)^{2/3}$
This is a REMOVABLE SINGULARITY in the scale factor.

Numerical result ($H = 0.1$):
 $t_{\text{reheat}} = 0.7457$
 $a(t_{\text{reheat}}) = 1.0774$ (both expressions match to 4 digits)

KEY: At $t = 0$, $a_{\text{inflation}}(0) = 1$, $a_{\text{matter}}(0) = 0$.
 $0^0 = 1$ vs $0^{2/3} = 0$. The $0/0$ is resolved by reheating.
The removable value $a(t_{\text{reheat}})$ determines T_{reheat} .

$\zeta(0) = -1/2$: The Empty Set Value

The functional equation:

$$\zeta(s) = 2^s \cdot \pi^{\{s-1\}} \cdot \sin(\pi s/2) \cdot \Gamma(1-s) \cdot \zeta(1-s)$$

At $s = 0$:

$$2^0 = 1$$

$$\pi^{\{-1\}} = 1/\pi$$

$$\sin(0) = 0$$

$$\Gamma(1) = 1$$

$$\zeta(1) = \text{infinity (mpmath raises 'ValueError: zeta(1) pole')}$$

$$\text{So: } \zeta(0) = 1 \cdot (1/\pi) \cdot 0 \cdot 1 \cdot \text{infinity}$$

$$= 0 \cdot \text{infinity}$$

$$= -1/2 \text{ (removable value)}$$

The pole at $s = 1$ is CONFIRMED by the code itself.

mpmath REFUSES to compute $\zeta(1)$. The $0/0$ is real.

Numerical verification ($s \rightarrow 0$):

$$s = 0.1000: \zeta(s) = -0.6030, \chi(s) \cdot \zeta(1-s) = -0.6030$$

$$s = 0.0100: \zeta(s) = -0.5093, \chi(s) \cdot \zeta(1-s) = -0.5093$$

$$s = 0.0010: \zeta(s) = -0.5009, \chi(s) \cdot \zeta(1-s) = -0.5009$$

$$s = 0.0001: \zeta(s) = -0.5001, \chi(s) \cdot \zeta(1-s) = -0.5001$$

As $s \rightarrow 0$: $\zeta(s) \rightarrow -1/2$.

The functional equation converges to the removable value.

BONUS: $\zeta(-1) = -1/12$

Another $0/0$: $1+2+3+\dots$ diverges, analytic continuation gives $-1/12$

$$0 \cdot \text{inf} = -1/12$$

BONUS: $\zeta(-2) = 0$ (trivial zero)

The $\sin(\pi s/2)$ factor gives zeros at $s = -2, -4, -6, \dots$

These are TRIVIAL zeros: $0 \cdot \text{finite} = 0$.

The Complete Picture

$0^0 = 1$: CREATION FROM NOTHING

The void has structure (empty function: exactly 1)
The origin is well-defined (power series survives)
The product diverges (Euler product at $s = 0$)
Cosmologically: $a(0) = 1$ (inflation, no Big Bang)

$0^x = 0$: ANNIHILATION AT THE ORIGIN

The origin destroys everything (algebra)
The limit is zero (calculus)
The Big Bang exists (Friedmann: $a(0) = 0$)
Cosmologically: $a(0) = 0$ (post-inflation, Big Bang)

THE TRANSITION: END OF INFLATION

$0^0 = 1 \rightarrow 0^x = 0$
The moment the universe 'chooses' to have a Big Bang
The reheating temperature T_{reheat} determines everything
At t_{reheat} : $\exp(H \cdot t) = (3t/2)^{2/3}$ (removable $0/0$)

$\zeta(0) = -1/2$: THE EMPTY SET VALUE

The removable value of $\sin(0) * \zeta(1) = 0 * \infty$
The zeta function 'knows' about the empty set
The prime spectrum begins at $-1/2$

$\zeta(-1) = -1/12$: THE SUM OF ALL INTEGERS

$0 * \infty = -1/12$ (the 'famous' regularization)

$\zeta(-2n) = 0$: THE TRIVIAL ZEROS

$0 * \text{finite} = 0$ (the sine factor annihilates)

THE $0/0$ UNIVERSE:

$0^0 = 1$ (creation) $\rightarrow 0^x = 0$ (annihilation)
The transition is the Big Bang (end of inflation)
The removable value is the initial condition
The zeta zeros are the harmonics of growth
The trivial zeros are the 'silent' modes ($0 * \text{finite} = 0$)
The non-trivial zeros are the 'active' modes ($0 * \infty = \text{nonzero}$)

Computed with mpmath (30 digits). All numerical results verified.